Retail Demand Forecasting & Inventory Optimization

Goal of the Project:
Predict how many units of each product will be sold in the future so that a store can **avoid running out of stock** or having too much leftover inventory.

Retailers often face challenges in predicting product demand, leading to stockouts or overstock, which results in lost revenue or increased costs.

This project aims to forecast daily product demand for a retail store, simulate inventory requirements, and optimize stock levels.

Using historical sales transactions, pricing, promotions, and seasonal trends, the project will provide actionable insights for business decisions and will be presented through interactive dashboards.

Data Cleaning Overview

Objective:
Prepare the BrightMart Retail dataset for analysis and modeling by handling missing values, outliers, formatting issues, and incorrect datatypes.

Missing Values:

- Item → filled with **'Unknown'**
- Price Per Unit → **category-wise mean**
- Quantity → **category-wise median**
- Revenue → recalculated using Price * Quantity * (1 - Promotion/100)
- Discount Applied → True/False based on Promotion

Column & Text Cleaning:

- Removed extra spaces, standardized **column names to Title Case**
- Cleaned categorical/text columns

Unnecessary Columns:

- Dropped Transaction ID & Customer ID

Datatype Corrections:

- Converted Transaction Date → **datetime**

Outlier Check:

- Revenue outliers identified but kept (high price/quantity purchases are realistic)

Duplicates:

- No duplicate rows found

Dataset is **clean, consistent, and ready for Feature Engineering & Modeling**

Feature Engineering

New Features Added:

- **Average Order Value (AOV)** Calculated as:

Revenue / Quantity

- **Promotion Impact** Quantifies the absolute discount amount:

(Price Per Unit × Quantity) − Revenue

- **Promotion Impact Per Unit**: Normalizes discount effect per item sold.
- **Time-Based Features**: Extracted Year, Month Number, Day, and Week Number from transaction date to support trend analysis.
- **Units Sold**: Derived directly from quantity purchased to measure sales volume.

Why These Features Matter

•Enable **better KPI tracking** (AOV, units sold, promotion effectiveness).
•Support **business decision-making** around pricing, promotions, and inventory planning.
•Prepare the dataset for **advanced analytics and predictive modeling**.

Feature Engineering & EDA

Insights:

- Revenue increases with promotions; impact varies by product category
- Some categories show stronger revenue response at lower discount levels
- Sales remain stable across weekdays and seasons, indicating consistent demand

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- One high-selling item labeled “Unknown” suggests data quality issues
- Customers use all payment methods evenly, with slight preference for digital wallets
- Online and in-store channels contribute nearly equal revenue, showing strong omnichannel behavior

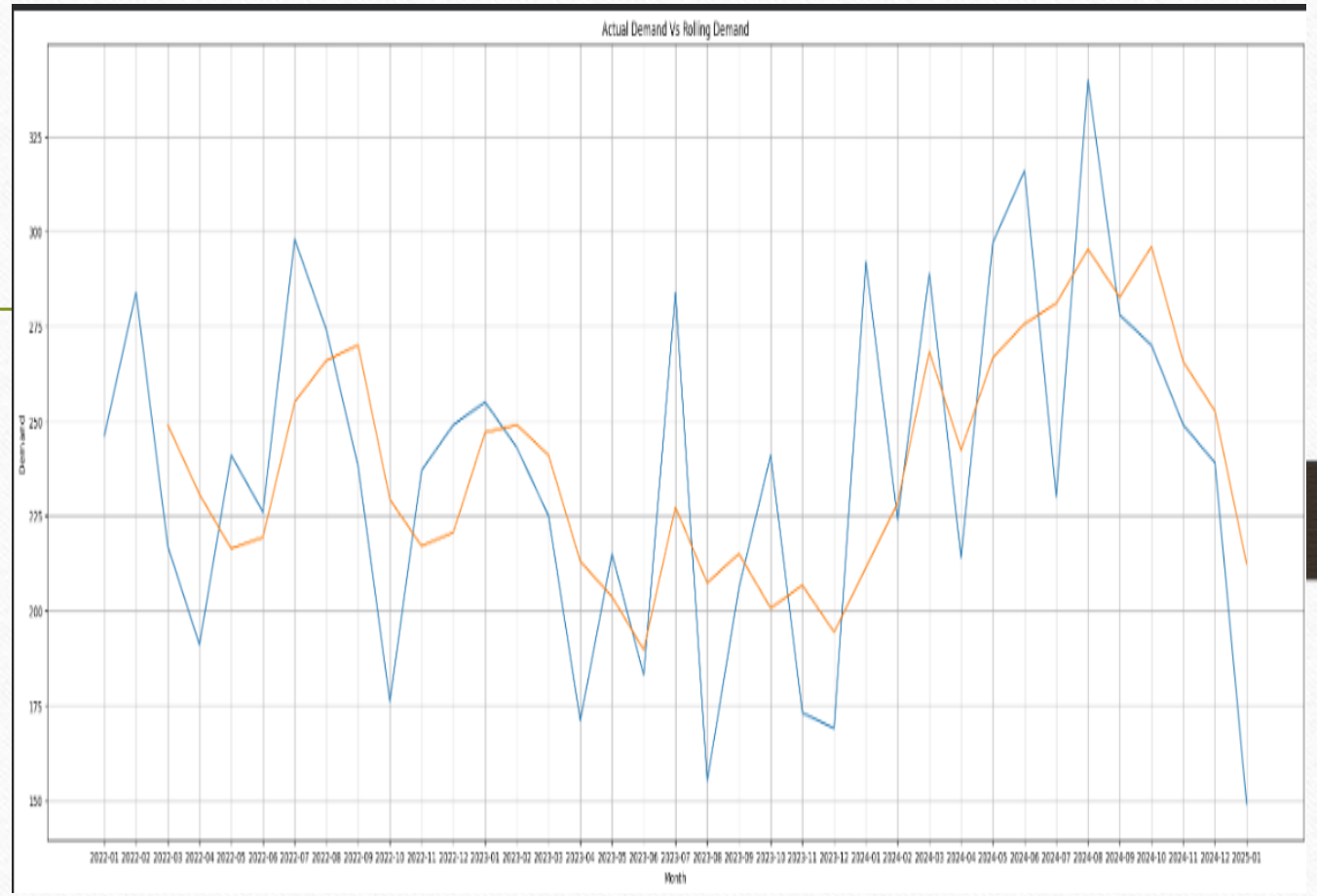
Feature Engineering & EDA

Business Recommendations:

- Apply targeted promotions by category instead of uniform discounts
- Validate margins before increasing discount depth
- Fix product tagging to improve forecasting and pricing decisions
- Promote digital wallet payments using incentives
- Maintain balanced investment across online and physical channels
- Identified key revenue drivers and seasonal demand patterns.
- Enabled data-driven promotion strategies.
- Created analytics-ready features aligned with real-world business KPIs.
- Established a strong foundation for forecasting and optimization models.

Demand Forecasting Approach

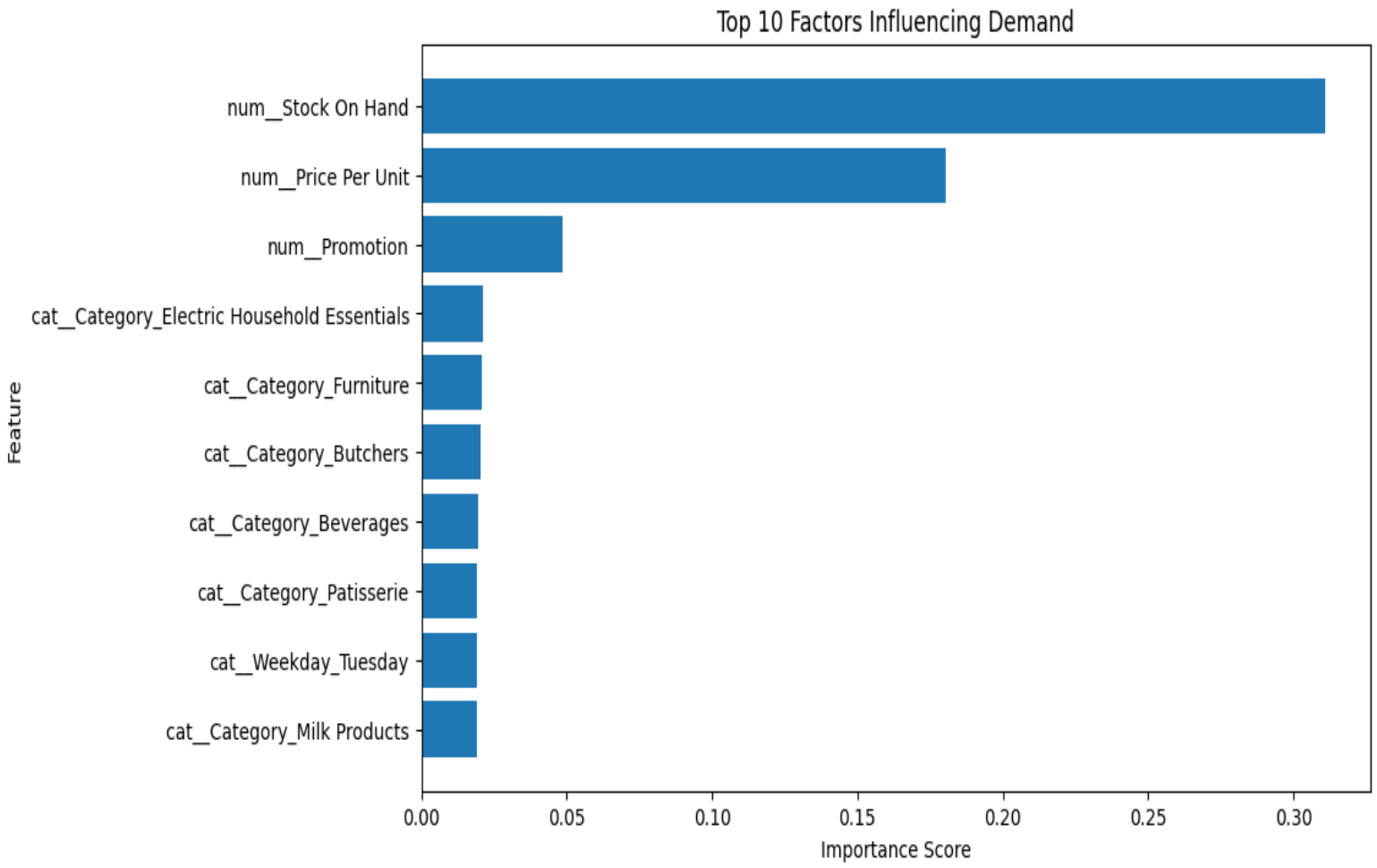
- Goal: Predict product demand (units sold) using historical data
- • Target variable: Quantity (units sold)
- • Model used: Random Forest Regression
- • Inputs:
- - Price
- - Promotions
- - Seasonality (Month, Weekday)
- - Inventory availability
- • Train-test split used time-based splitting (no shuffling)
- • Evaluation metrics: MAE, RMSE



We used a machine learning model to predict how many units will be sold, not revenue.
This allows inventory decisions to be made directly from forecasts.

What Drives Demand? (Model Explainability)

- Used model-based feature importance (Random Forest)
- Focused on interpretability for business users
- Top demand drivers:
- Stock on Hand
- Price per Unit
- Promotion Intensity
- Product Category
- Day of Week



Stock availability is the strongest driver of realized demand.
Promotions and price matter, but only when inventory is available.
Future Scope: SHAP-based explainability for deeper insights

Inventory Optimization Framework

- Forecasted demand converted into inventory decisions
- Key components:

Average Daily Demand

Lead Time (7 days)

Safety Stock (95% service level)

Reorder Point (ROP)

- Formula:

ROP = (Avg Daily Demand × Lead Time) + Safety Stock

Instead of stopping at forecasting, we convert predictions into clear reorder rules so the business knows when to place orders.

Demand

Safety Stock

Reorder Point

Order Decision

Cost Impact Analysis

- Compared two costs:

Stockout Cost (lost sales)

Holding Cost (extra inventory)

- Simulated promotion-driven demand spikes
- Key finding:

Stockout cost is consistently higher than holding cost

From a financial perspective, losing sales hurts more than carrying a small buffer of extra inventory

Cost Impact Simulation

Estimated Stockout Cost	Holding Cost
$4	$56

Interactive Decision-Making Dashboard

- Live Streamlit dashboard
- Enables category-level analysis
- Shows:

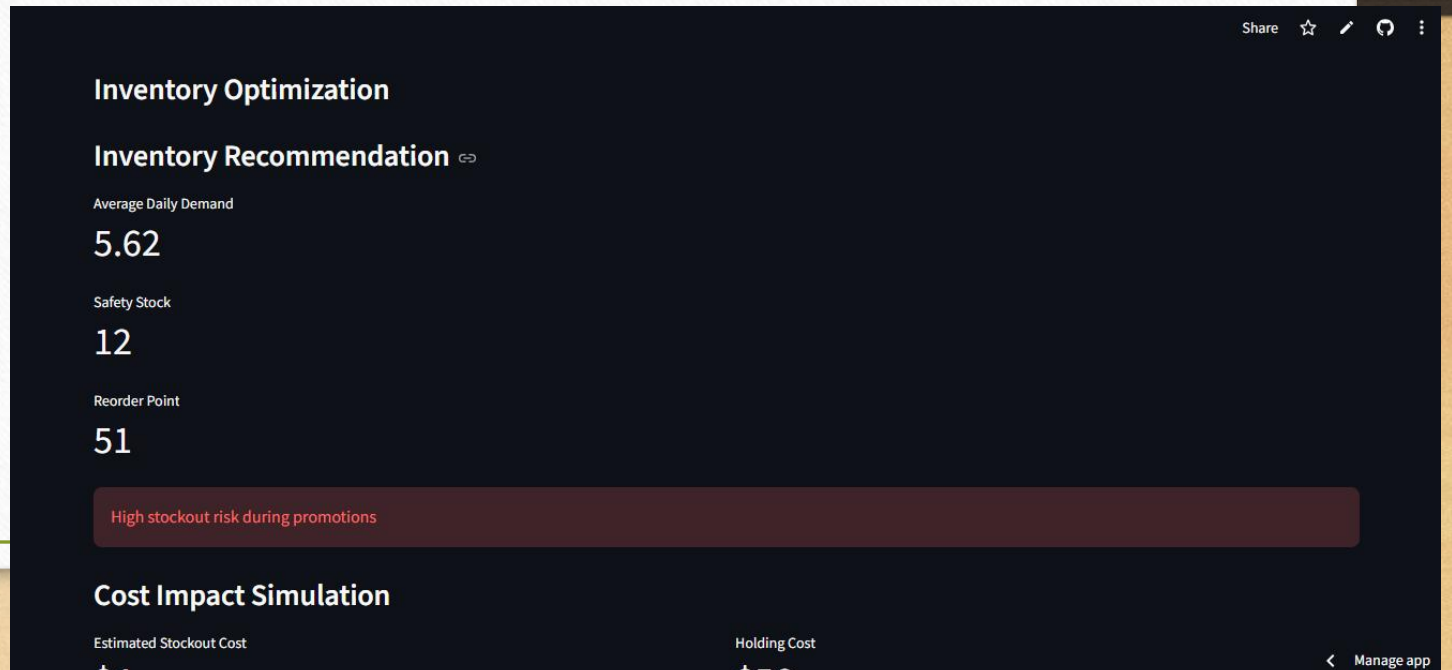
Demand trends

Promotion impact

Inventory recommendations

Cost risk comparison

Live App: https://retail-demand-inventory-dashboard-gjn7qnutxhgeqn6kvpsl2f.streamlit.app/



Key Business Recommendations

- Maintain safety stock across all categories
- Increase inventory before running promotions
- Avoid promotions without inventory readiness
- Accept moderate holding cost to prevent revenue loss
- Use data-driven reorder points instead of intuition
- Inventory availability, not discounts alone, determines realized sales.